

CSE428 Lab Project Instructions

Title: Brain Tumor MRI Image Classification and Segmentation

Marks Distribution (out of 15):

1. Attendance - 1%
2. Presentation - 2%
3. Demonstration - 6%
4. Individual Viva - 6%

Presentation Slide Contents:

- Provide a **brief dataset description** along with its **training/testing/validation split** and **class balance and distribution (in bar chart)**.
- Define loss functions, add training/testing/validation **loss/accuracy vs epoch curves** etc.
- Mention the **hyperparameters** - learning rate, optimizers, batch size, no. of epochs, regularization parameters etc.
- Present key **experimental results** such as **accuracy, precision, recall, F1-score** and the **confusion matrix** for classification task and **dice-coefficient, mIoU score and pixel accuracy** etc. for segmentation task for all sets - training, testing & validation (if used).
- Include **example images** of both **good (correct)** and **bad (incorrect)** predictions (both segmentation & classification).

Demonstration:

For the **demonstration part**, we will provide you with **some random images** on which you have to **run your model and get the predictions**. Follow the instructions given in minimum expectations to code this part up. Do note that we might ask you to execute any block of code or simply ask you the function of the code block. Failure to do so will result in **mark deduction** from this segment under the GPT penalty.

Viva:

Finally in the viva, we will ask you questions about **your understanding of the project**. Even if you haven't worked on one part of the project, you may still be asked from that part. **So it's your responsibility to understand the project properly.**

Project Presentation will take place on 29th December 2025 (Monday) during scheduled lab class time.

Tasks to do:

1. Dataset Review & Preprocessing

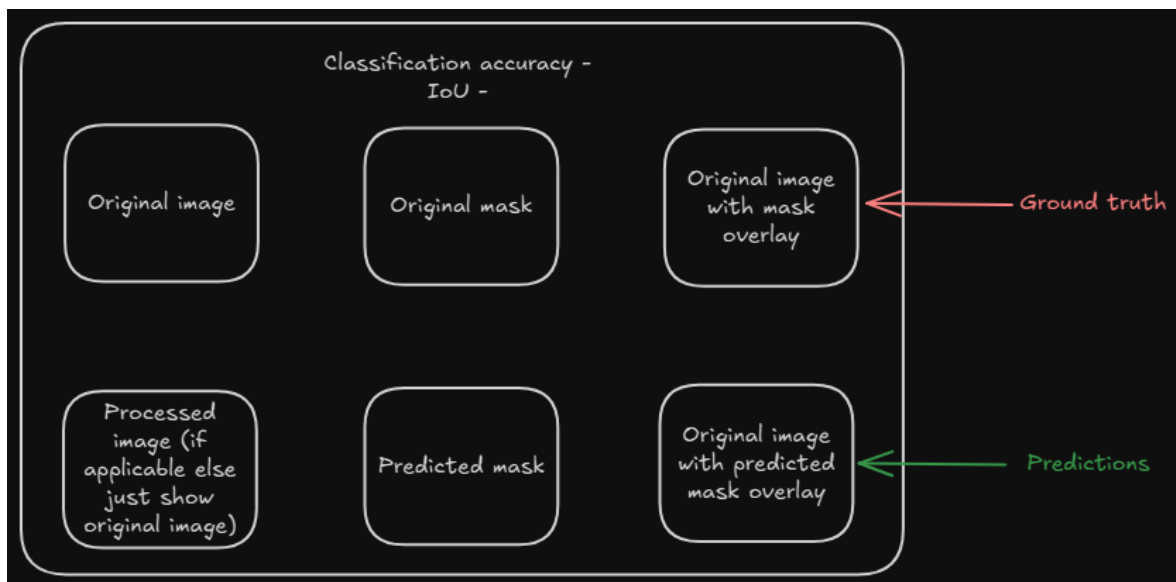
Dataset Link: [BRISC 2025 Dataset](#)

- **Understand the data:** Review dataset documentation to learn acquisition and annotation details.
 - **Explore files:** Check image formats, folder structure, and annotations.
 - **Preprocessing pipeline:**
 - Identify a common pattern for preprocessing instead of processing each image individually.
 - Resize and normalize pixel values.
 - Apply denoising or morphological filters if needed.
 - Ensure lesion masks and labels are correctly parsed and aligned.
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2. Model Design

Base Architecture

- **Prepare a U-Net Architecture** ([Paper](#)) to segment the data.
- Attach a **classifier head** to the encoder output and classify the data.
- Prepare a code that takes any image as input and outputs:



Updated Architecture

- Update the implemented **U-Net to Attention U-Net** ([Paper](#)) and rerun **segmentation training**.
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3. Training Strategy

- You may train segmentation and classification heads **separately or jointly**.
 - If you compare results from separate vs. joint training with analysis, you can earn bonus marks.
 - Use **appropriate loss functions**:
 - Segmentation: Dice-loss or cross-entropy.
 - Classification: Cross-entropy.
 - Use **optimizers, learning-rate scheduling, early stopping**, and **proper batch sizes** to maximize performance.
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4. Evaluation Metrics

Segmentation:

- Mean Intersection over Union (mIoU)
- Dice coefficient
- Pixel accuracy

Classification:

- Accuracy
 - Precision, Recall, F1-score
 - Confusion Matrix
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5. Open-Ended Design Choices

- Any classifier architecture may be used — choose for best accuracy.
 - Preprocessing is flexible; you can find a common pattern and apply a preprocessing pipeline — your goal is to achieve the **best possible results**.
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6. Bonus Tasks (Optional)

Bonus marks are awarded at the faculty's discretion. Incomplete bonus work gets **0 marks**. You may attempt multiple tasks:

1. Compare performance when training segmentation and classification heads separately vs. jointly. *(Easy)*
 2. Try different classifier architectures and analyze performance differences. *(Easy)*
Perform hyperparameter tuning (e.g., optimizers, learning rates) with compiled results. *(Medium)*
 3. Replace the U-Net decoder with **EfficientDet's BiFPN** ([Paper](#)) for multi-scale feature fusion. *(High)*
 4. Propose and get approval for any novel bonus idea.
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